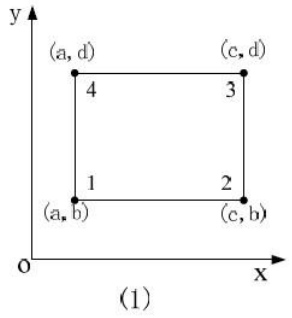
第 9 讲作业：

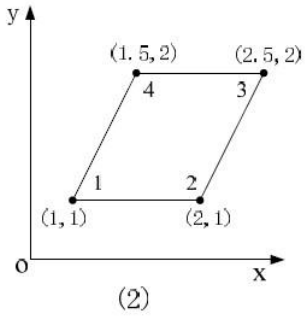
一、图示为四结点四边形等参数单元的子单元，试给出：

1、单元的几何模式；

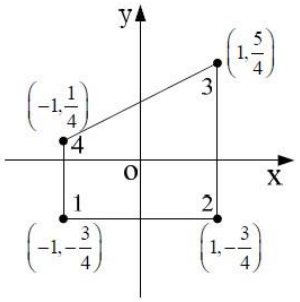
2、单元的Jacobian矩阵 $[J]$和$|J|$，并讨论是否满足等参变换的必要条件。



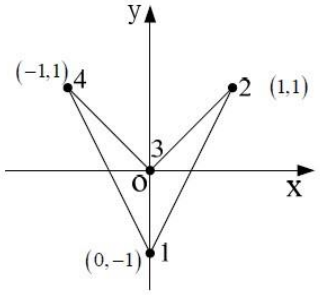
(1)



(2)



(3)

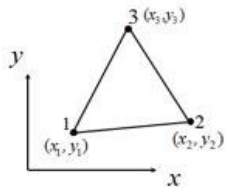


(4)

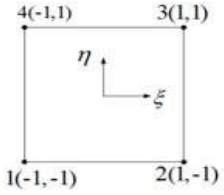
二、图示为三结点三角形等参单元（子单元为四结点四边形等参元的退化单元（结点 1 和 4 合并）），试给出：

1、单元的几何模式和位移模式；

2、单元的Jacobin矩阵 $[J]$ 和 $|J|$ ，并讨论是否满足等参变换的必要条件。



子单元



母单元

三、

Develop an appropriate mapping to map the quadrilateral shown in Figure 6.42 to the 2×2 square master area. Compute the determinant of the Jacobian matrix and using it demonstrate that the mapping is good. The coordinates of the key points are as follows.

	x	y
1	8.	0.
2	5.65685	5.65685
3	0.	8.
4	0	3
5	2	0

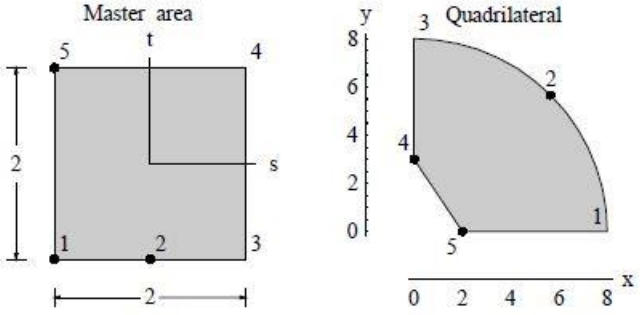


Figure 6.42.